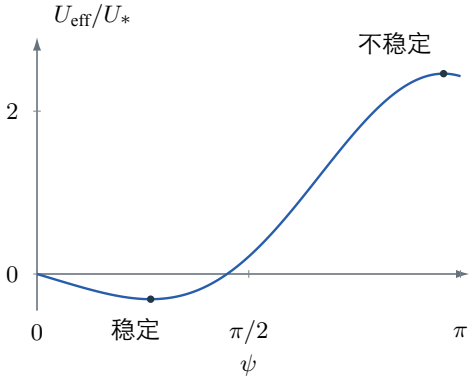
M5-166a　磁链、有效势与滚动惯量



$$LI + 2Brh\cos\psi = 2Brh$$
$$I = 2Brh(1-\cos\psi)/L$$

$$U_{\text{eff}} = U_*(1-\cos\psi)^2$$
$$-mgr\sin\alpha\,\psi,\quad U_* = 2B^2r^2h^2/L$$

$$J_{\text{eff}} = 3mr^2/2$$
$$\omega_{\text{s}}^2 = U''_{\text{eff}}/J_{\text{eff}}$$

左图仅示意 $\beta = mgr\sin\alpha/U_* = 0.5$；极小点稳定，极大点不稳定，不代表题设给了该数值。